# Fundamentals of Computer Design

(Computer Architecture: Chapter 1 )

Yanyan Shen

Department of Computer Science and Engineering

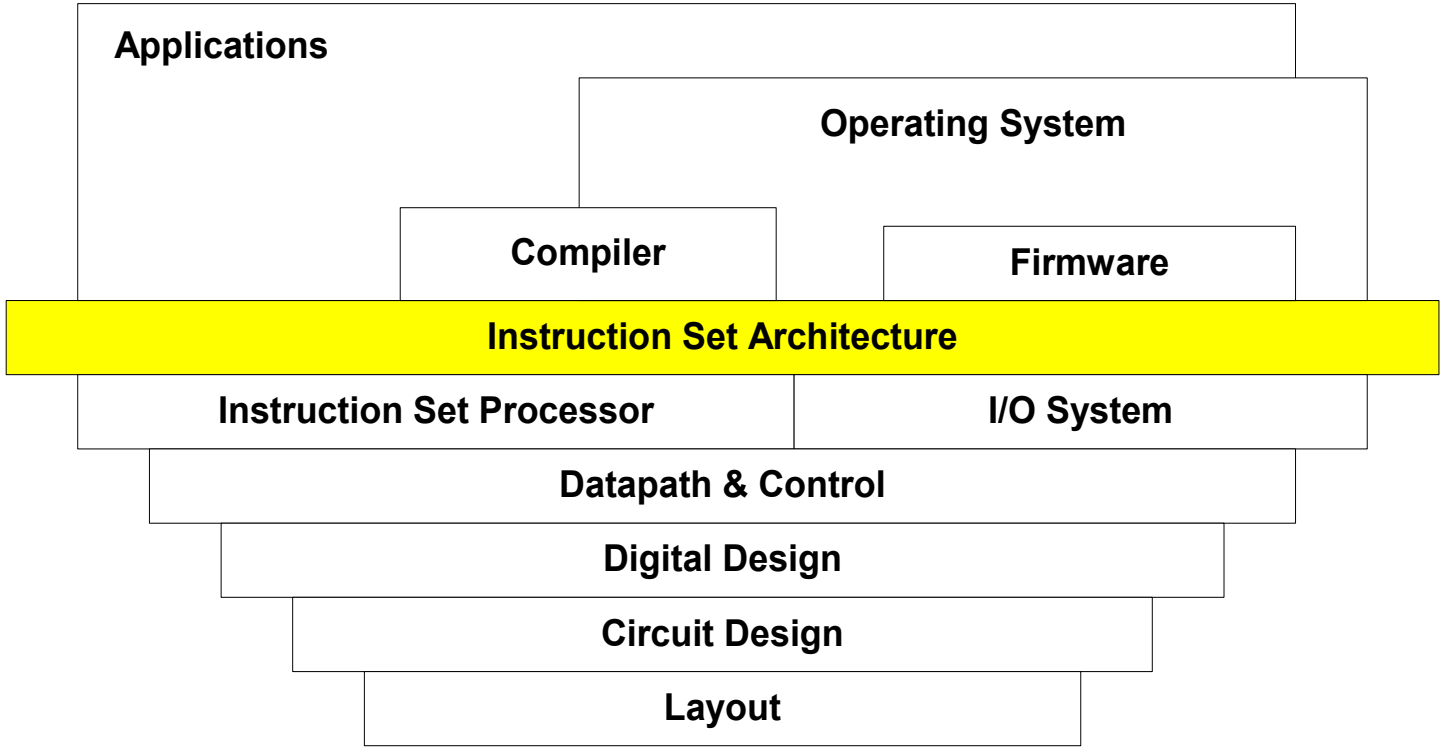
# Agenda

- Introduction
- Classes of Computers
- 1.3 Defining Computer Architecture
- Trends in Technology
- Trends in Power and Energy in ICs
- Trends in Cost
- Dependability
- Measuring Performance
- Quantitative Principles

# Defining Computer Architecture

- "Old" view of computer architecture:
  - Instruction Set Architecture (ISA) design
  - i.e., decisions regarding:
    - registers, memory addressing, addressing modes, instruction operands, available operations, control flow instructions, instruction encoding
- The ISA is roughly the same as the programming model of a processor as seen by an assembly language programmer or compiler writer.

# Instruction Set Architecture




MIPS, RISC-V
Load-Store ISA (不能直接对内存地址做操作，需要将数据load到寄存器操作)
指令简单、指令较长
简单就是好！

x86 ISA: Register Memory ISA
可以直接对内存操作。

S/W and H/W consists of hierarchical layers of abstraction, each hides details of lower layers from the above layer

**The instruction set arch.** abstracts the H/W and S/W interface and allows many implementation of varying cost and performance to run the same S/W

# Contents of Instruction Set Architecture

- Registers
- Class of ISA
- Memory addressing
  - Byte addressing

内存可以看作一个大数组，以二进制表示。
一般我们用16进制写，但实际机器采用2进制
一个内存字节对应一个byte

- Addressing modes
- Types and sizes of operands
- Operations
- Control flow instructions
- Encoding an ISA
  - Fixed length and variable length

# Addressing Modes

| Example Instruction | Meaning | When used |
|---|---|---|
| Register | Add R4,R3 | R4 <- R4 + R3 |
| Immediate（常数/立即数） | Add R4, #3 | R4 <- R4 + 3 |
| Displacement(寄存器相对寻址) | Add R4, 100(R1) | R4 <- R4 + M[100+R1] |
| Register deferred (寄存器间接寻址) | Add R4,(R1) | R4 <- R4 + M[R1] |
| Indexed | Add R3, (R1 + R2) | R3 <- R3 + M[R1+R2] |
| Direct | Add R1, (1001) | R1 <- R1 + M[1001] |
| Memory deferred | Add R1, @(R3) | R1 <- R1 + M[M[R3]] |
| Auto-increment | Add R1, (R2)+ | R1 <- R1 +M[R2]<br>R2 <- R2 + d |
| Auto-decrement | Add R1,-(R2) | R2 <-R2-d<br>R1 <- R1 + M[R2] |
| Scaled | Add R1, 100(R2)[R3] | R1<-R1+M[100+R2+R3*d] |

# Example ISAs

- ARM
  - ARMv1-v7
  - ARMv7 extensions
    - Thumb-2
    - NEON – media acceleration technology[8]
    - VFP v3
- HP
  - HP 2100
  - PA-RISC
- IBM
  - IBM 8100
  - IBM Series
  - Power Architecture
    - POWER
    - PowerPC
    - PowerPC AS
- MIPS

- Intel
  - IA-64
  - X86
  - x86 extensions
    - FPU (x87) – Floating-point-unit (FPU) instructions
    - MMX – MMX SIMD instructions
    - MMX Extended – extended MMX SIMD instructions
    - SSE – streaming SIMD extensions (SSE) instructions (70 instructions)
    - SSE2 – streaming SIMD extensions 2 instructions (144 new instructions)
    - SSE3 – streaming SIMD extensions 3 instructions (13 new instructions)
    - SSSE3 – supplemental streaming SIMD extensions (16 instructions)
    - SSE4.1 – streaming SIMD extensions 4, Penryn subset (47 instructions)
    - SSE4.2 – streaming SIMD extensions 4, Nehalem subset (7 instructions)
    - SSE4 – All streaming SIMD extensions 4 instructions (both SSE4.1 and SSE4.2)
    - SSE4a – streaming SIMD extensions 4a (AMD)
    - SSE5 – streaming SIMD extensions 5 (170 instructions)
    - XSAVE – XSAVE instructions
    - AVX – advanced vector extensions instructions
    - FMA – fused multiply-add instructions
    - AES – Advanced Encryption Standard instructions
    - CLMUL – Carry-less mtiply (PCLMULQDQ) instruction
    - 3DNow![citation needed] – 3DNow! instructions (21 instructions)
    - 3DNow! Extended – extended 3DNow! instructions (5 instructions)
    - Cyrix – Cyrix-specific instructions
    - AMD – AMD-specific instructions (older than K6)
    - SMM – System management mode instructions
    - SVM – Secure virtual machine instructions
    - PadLock – VIA PadLock instructions

# MIPS Instruction Fields

- MIPS fields are given names to make them easier to refer to

指令被拆分成不同的域(filed)进行编码

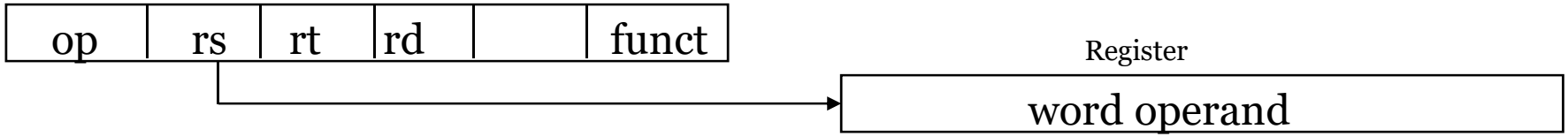
| op | rs | rt | rd | shamt | funct |
|---|---|---|---|---|---|

| | | | |
|---|---|---|---|
| op | 6-bits | opcode that specifies the operation | 操作编号 |
| rs | 5-bits | register file address of the first source operand | |
| rt | 5-bits | register file address of the second source operand | |
| rd | 5-bits | register file address of the result's destination | |
| shamt | 5-bits | shift amount (for shift instructions) | 偏移，位操作 |
| funct | 6-bits | function code augmenting the opcode | 补充，与op filed一起使用 |

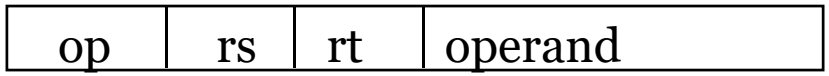
# MIPS Addressing Modes Examples

## 1. Register addressing

5bit恰巧对应32个寄存器的编号



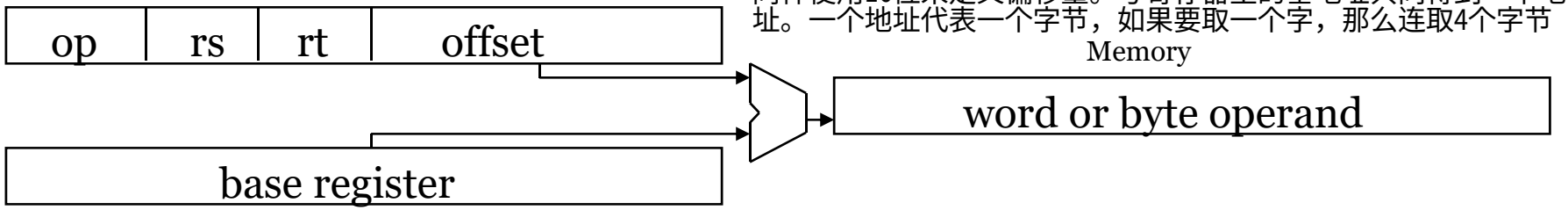

## 2. Immediate addressing




用rd，sm，funct三个域来定义立即数。
所以这个常数的范围是-2^15~2^15-1
日常使用的大部分常数在这个范围内

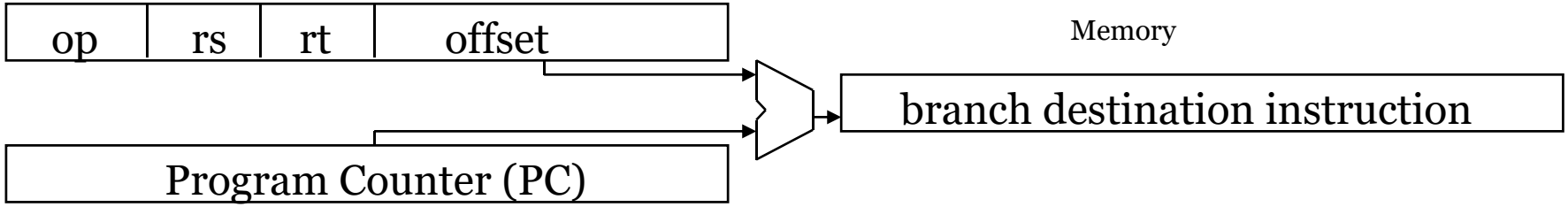
## 3. Base (displacement) addressing

同样使用16位来定义偏移量。与寄存器里的基地址共同得到一个地址。一个地址代表一个字节，如果要取一个字，那么连取4个字节



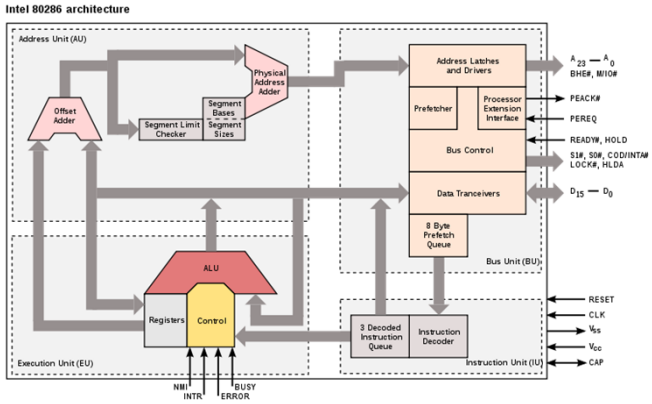

## 4. PC-relative addressing

PC相对寻址：默认将PC存储的地址作为基地址，用于跳转等。



# Microarchitecture

- **microarchitecture**, also called **computer organization**, is the way a given ISA is implemented on a processor.
  - A given ISA may be implemented with different microarchitectures.
- The microarchitecture includes the constituent parts of the processor and how these interconnect and interoperate to implement the ISA.




**Intel i80286 microarchitecture**

# Microarchitectural Concepts

- Instruction pipelining
- Hierarchical memory organization
- Cache
- Cache coherence
- Branch prediction
- Superscalar
- Out-of-order execution
- Register renaming
- Multiprocessing and multithreading

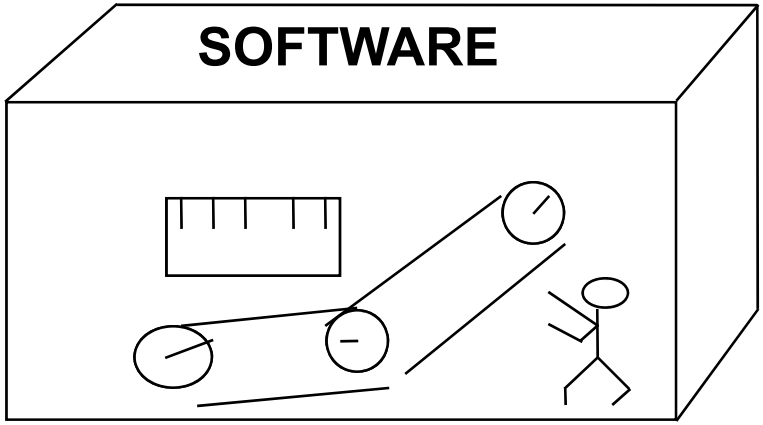
# Computer Architecture in General

When we construct a building numerous practical considerations need to be taken into account:

- Available materials
- Worker skills
- Budget
- Space

Similarly, Computer Architecture is about working within constraints:

- What will the market buy?
- Cost/Performance
- Tradeoffs in materials and processes



# Genuine Computer Architecture

- **Genuine definition**: *Designing the organization and hardware to meet goals and functional requirements*
- The implementation of a computer involves two components
  - 1) **organization** (or microarchitecture)
  - 2) **hardware**: detailed logic design and packaging
- In this course, **computer architecture** covers all the three:
  - (1) ***ISA***,
  - (2) ***microarchitecture*** *or* ***organization***, and
  - (3) ***hardware***

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# Rapid Changes in Technology

- If an ISA is successful, it should survive rapid changes in *implementation technology*

  为了兼容性考虑，优秀的 ISA 需要绝对的稳定性，在硬件变革的情况下也能保证不变。

- The designer of a computer should be aware of such rapid changes

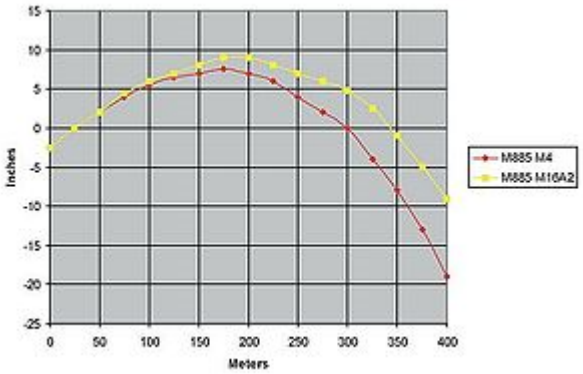
- **Five implementation technologies** are critical to model implementations of computer design

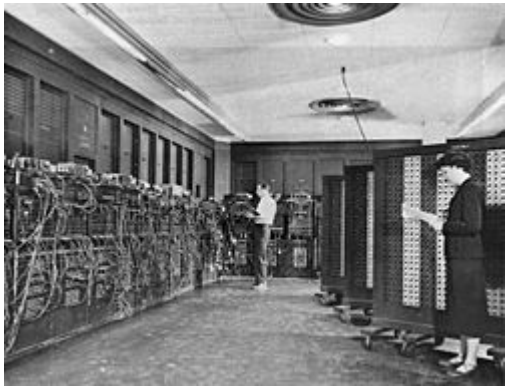
# Five Critical Implementation Techs

| | |
|---|---|
| Integrated circuit logic technology | • As Moore's law, the transistors count doubles every 18-24 months |
| Semiconductor DRAM | • The capacity per DRAM chip doubles every two to three years |
| Semiconductor Flash | • Capacity doubles roughly every two years |
| Magnetic disk technology | • Capacity doubles every three years |
| Network technology | • Bandwidth increases quickly |

# ENIAC - background

- Electronic Numerical Computer
- Eckert and Mauchly, University of Pennsylvania
- Purpose: *Trajectory tables for weapons*
- Started 1943, Finished 1946
  - Too late for war effort
  - Used until 1955



John Mauchly (1907-1980) &
J. Presper Eckert (1919-1995)

# ENIAC - details

- Decimal (not binary)
- 20 accumulators of 10 digits
- Programmed manually by switches
- 18,000 **vacuum tubes**
- 30 tons
- 15,000 square feet
- 140 kW power consumption
- 5,000 additions per second

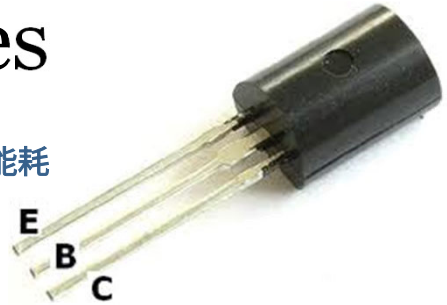
初代设计是十进制计算机（使用BCD表示）
有二十个十进制加法器
通过开关电路手动编程
18000真空管，30吨，15000平方英尺
140kw能耗，每秒5000次加法

| Straight ring/Overbeck counter | | | | |
|---|---|---|---|---|
| State | Q0 | Q1 | Q2 | Q3 |
| 0 | 1 | 0 | 0 | 0 |
| 1 | 0 | 1 | 0 | 0 |
| 2 | 0 | 0 | 1 | 0 |
| 3 | 0 | 0 | 0 | 1 |
| 0 | 1 | 0 | 0 | 0 |
| 1 | 0 | 1 | 0 | 0 |
| 2 | 0 | 0 | 1 | 0 |
| 3 | 0 | 0 | 0 | 1 |
| 0 | 1 | 0 | 0 | 0 |

# Transistors – the 2nd generation

- Replaced vacuum tubes
- Advantages
  - Smaller
  - Cheaper
  - Less heat dissipation
- Solid state device
- Made from Silicon (Sand)
- Invented 1947 at **Bell Labs**
- William Shockley et al.

使用晶体管，
更便宜、更小、更少的能耗
固体态设备



First working transisto

John Bardeen, William Shockley and Walter Brattain at Bell Labs, 1948.

# Transistor Based Computers

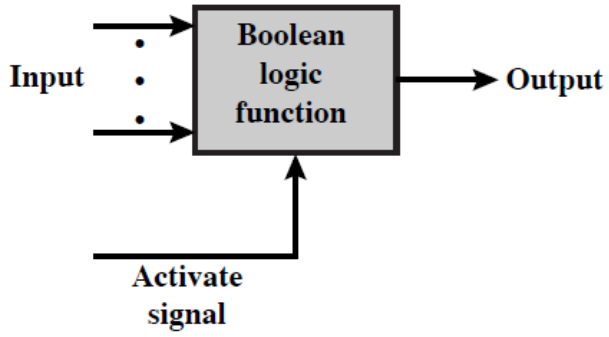
- Second generation machines
- NCR & RCA produced small transistor machines
- IBM
  - Produced IBM 7000
- DEC - 1957
  - Produced PDP-1

IBM 7000

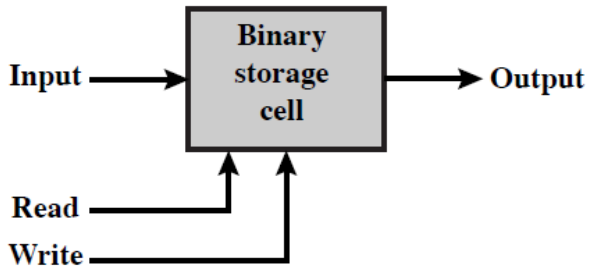
PDP-1

# Microelectronics – 3rd Generation

- Literally - "small electronics"
- A computer is made up of gates, memory cells and interconnections
- These can be manufactured on a **<u>semiconductor</u>**
  - e.g. silicon wafer




(a) Gate




(b) Memory cell

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# Two Metrics: Bandwidth and Latency

## Bandwidth or throughput 带宽或吞吐量

- *Total work done in a given time*
- 10,000-25,000X improvement for processors
- 300-1200X improvement for memory and disks

带宽增速相比延迟更快

## Latency or response time 延迟或响应时间

- *Time between start and completion of an event*
- 30-80X improvement for processors
- 6-8X improvement for memory and disks

# Fundamentals

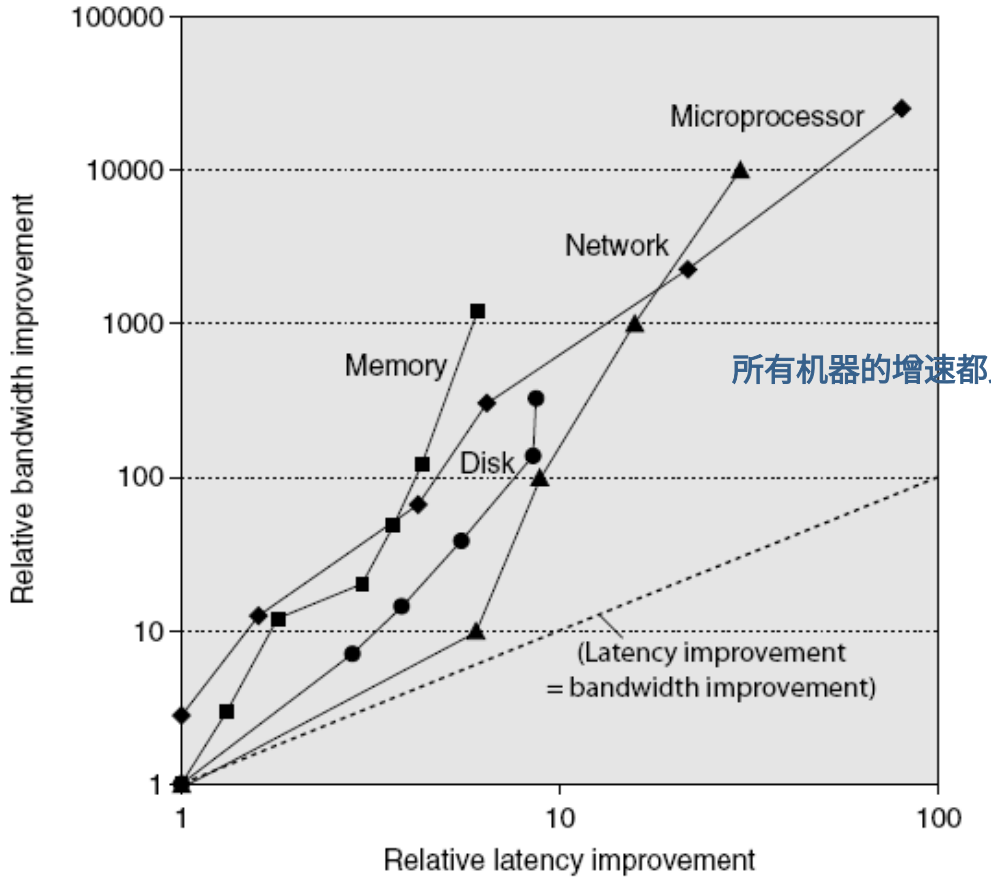
| Microprocessor | 16-bit address/ bus, microcoded | 32-bit address/ bus, microcoded | 5-stage pipeline, on-chip I & D caches, FPU | 2-way superscalar, 64-bit bus | Out-of-order 3-way superscalar | Out-of-order superpipelined, on-chip L2 cache | Multicore OOO 4-way on chip L3 cache, Turbo |
|---|---|---|---|---|---|---|---|
| Product | Intel 80286 | Intel 80386 | Intel 80486 | Intel Pentium | Intel Pentium Pro | Intel Pentium 4 | Intel Core i7 |
| Year | 1982 | 1985 | 1989 | 1993 | 1997 | 2001 | 2010 |
| Die size ($mm^2$) | 47 | 43 | 81 | 90 | 308 | 217 | 240 |
| Transistors | 134,000 | 275,000 | 1,200,000 | 3,100,000 | 5,500,000 | 42,000,000 | 1,170,000,000 |
| Processors/chip | 1 | 1 | 1 | 1 | 1 | 1 | 4 |
| Pins | 68 | 132 | 168 | 273 | 387 | 423 | 1366 |
| Latency (clocks) | 6 | 5 | 5 | 5 | 10 | 22 | 14 |
| Bus width (bits) | 16 | 32 | 32 | 64 | 64 | 64 | 196 |
| Clock rate (MHz) | 12.5 | 16 | 25 | 66 | 200 | 1500 | 3333 |
| Bandwidth (MIPS) | 2 | 6 | 25 | 132 | 600 | 4500 | 50,000 |
| Latency (ns) | 320 | 313 | 200 | 76 | 50 | 15 | 4 |
| Memory module | DRAM | Page mode DRAM | Fast page mode DRAM | Fast page mode DRAM | Synchronous DRAM | Double data rate SDRAM | DDR3 SDRAM |
| Module width (bits) | 16 | 16 | 32 | 64 | 64 | 64 | 64 |
| Year | 1980 | 1983 | 1986 | 1993 | 1997 | 2000 | 2010 |
| Mbits/DRAM chip | 0.06 | 0.25 | 1 | 16 | 64 | 256 | 2048 |
| Die size ($mm^2$) | 35 | 45 | 70 | 130 | 170 | 204 | 50 |
| Pins/DRAM chip | 16 | 16 | 18 | 20 | 54 | 66 | 134 |
| Bandwidth (MBytes/s) | 13 | 40 | 160 | 267 | 640 | 1600 | 16,000 |
| Latency (ns) | 225 | 170 | 125 | 75 | 62 | 52 | 37 |
| Local area network | Ethernet | Fast Ethernet | Gigabit Ethernet | 10 Gigabit Ethernet | 100 Gigabit Ethernet | | |
| IEEE standard | 802.3 | 803.3u | 802.3ab | 802.3ac | 802.3ba | | |
| Year | 1978 | 1995 | 1999 | 2003 | 2010 | | |
| Bandwidth (Mbits/sec) | 10 | 100 | 1000 | 10,000 | 100,000 | | |
| Latency (μsec) | 3000 | 500 | 340 | 190 | 100 | | |
| Hard disk | 3600 RPM | 5400 RPM | 7200 RPM | 10,000 RPM | 15,000 RPM | 15,000 RPM | |
| Product | CDC WrenI 94145-36 | Seagate ST41600 | Seagate ST15150 | Seagate ST39102 | Seagate ST373453 | Seagate ST3600057 | |
| Year | 1983 | 1990 | 1994 | 1998 | 2003 | 2010 | |
| Capacity (GB) | 0.03 | 1.4 | 4.3 | 9.1 | 73.4 | 600 | |
| Disk form factor | 5.25 inch | 5.25 inch | 3.5 inch | 3.5 inch | 3.5 inch | 3.5 inch | |
| Media diameter | 5.25 inch | 5.25 inch | 3.5 inch | 3.0 inch | 2.5 inch | 2.5 inch | |
| Interface | ST-412 | SCSI | SCSI | SCSI | SCSI | SAS | |
| Bandwidth (MBytes/s) | 0.6 | 4 | 9 | 24 | 86 | 204 | |
| Latency (ms) | 48.3 | 17.1 | 12.7 | 8.8 | 5.7 | 3.6 | |

**Milestones for**

1) microprocessors,
2) memory,
3) networks, and
4) disks

# Bandwidth and Latency

- Latency improved 6x to 80x
- Throughput improved 300x to 25,000x
- **Bandwidth has outpaced latency!** 可以靠并发来提升，所以增速更快




所有机器的增速都比延迟更快

Log-log plot of bandwidth and latency milestones

# Feature Size and Its Impacts

**Feature size** is the minimum size of transistor or wire in x or y dimension

- **Integration density** scales quadratically with feature size
  - 10 microns in 1971 to .032 microns in 2011
- **Wire delay** does not scale well with feature size!
  - Wire delay is proportional to the product of <u>resistance</u> and <u>capacitance</u>

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# Power and Energy

**Problem: Get power in, get power or heat out**

- ***Thermal Design Power (TDP)*** 散热设计功耗
  - Characterizes sustained power consumption
  - Used as target for power supply and cooling system
  - Lower than peak power, higher than average power consumption

    比平常的功耗高，比峰值低
    在这种峰值情况下，会用一些其他的设计降低功耗

- **Clock rate** can be reduced dynamically to limit power consumption

  时钟周期会变得更慢来适应散热功耗

# Dynamic Energy and Power

- ***Dynamic energy per transistor*** 能量估计：1/2CV^2
  - Used for a transistor switching from 0 -> 1 or 1 -> 0
  - ½ x Capacitive load x Voltage$^2$ 每一个晶体管的跳变都会带来功耗，即动态能耗
  - The capacitive load is a function of *number of transistors* connected to output and *the technology* which determines the capacitance of wires and transistors
- ***Dynamic power*** *per transistor*
  - ½ x Capacitive load x Voltage$^2$ x Frequency switched

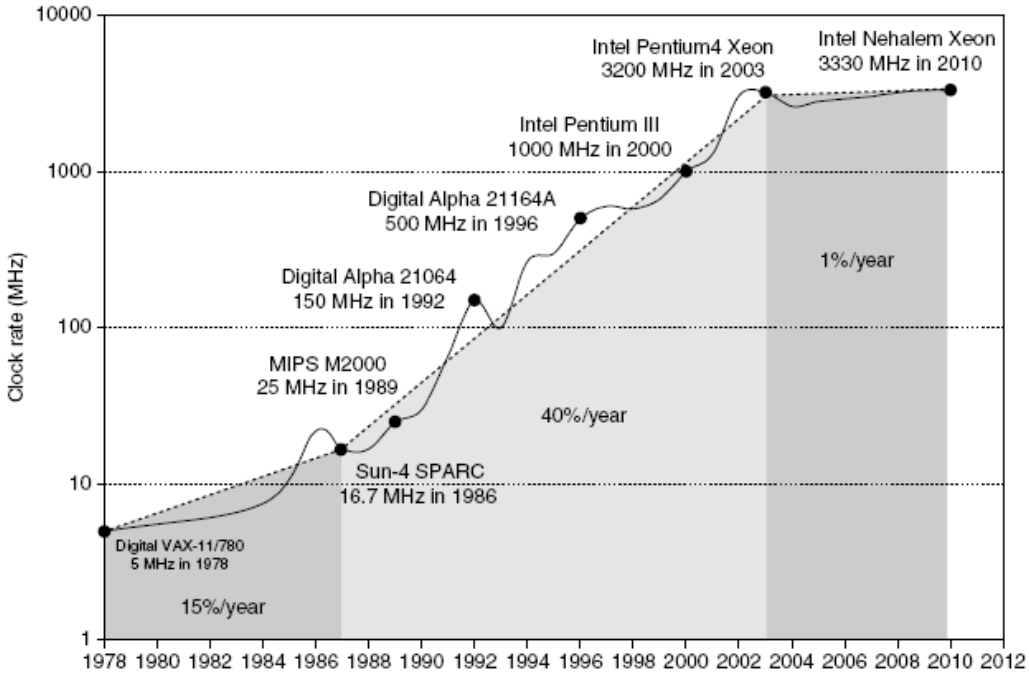
由于设计和硬件技术上升，电容两端电压下降。所以功耗的增加没有频率增加的快

- **Voltage is the key**
  - The voltage of processors has become lower and lower

# Power

- **Intel 80386** consumed ~ 2 W
- 3.3 GHz **Intel Core i7** consumes **130 W**
- Heat must be dissipated from 1.5 x 1.5 cm chip

- **This is the limit of what can be cooled by air**



# Power vs. Energy

- **Power** is defined as the energy consumed in a unit time

更多情况要看能耗而非功耗。能耗反映总体水平，不受速度影响。它能反映电费

- Question: Which metric should we use to compare processors: power or energy?

- It is better to use energy used for a specific task
  - Energy is tied to the a specific task and the time required for that task

# Techniques for Reducing Power

减少功耗的方法：改变时钟周期的模式、改变内存的方式、减少核的运转数量.....

| | |
|---|---|
| Turn off the clock | • Turn off the clock of inactive modules |
| Dynamic Voltage-Frequency Scaling | • In periods of low activity, operate at lower frequency |
| Low power state for DRAM, disks | • In low power states, accesses are not allowed |
| Overclocking, turning off cores | • To turn off other cores and just run on a few cores |

# Static Power

- **Cause: The leakage current flows even when a transistor is off**

  静态耗电：晶体管数量越多，掉电越快。只要供电就会有静态功耗（其实是漏电）

- $\text{Power}_{\text{static}} = \text{Current}_{\text{static}} \times \text{Voltage}$
  - Scales with number of transistors

- To reduce static power: ***power gating***, <u>turning off the power supply</u>